INTRODUCTION TO HEALTH ECONOMICS

Quiz 2

Time: 45 minutes Maximum Marks: 10

Name: ________________________________ **Roll No:** ________________________________

Question: An individual faces a probability of falling sick of $\mathbf{p = 0.5}$. Her utility function is $\mathbf{U(I) = 1000I - I^2}$, where I is income. Assume: income when healthy = 1000; income when sick = 0. She has two insurance options:

- **Contract A:** Full insurance with a premium of 600.
- **Contract B:** Fair but partial insurance with payout q and premium = $p \times q$.

At what value of q is this individual indifferent between the two contracts? Provide a detailed explanation.